

Logical Database Design Normalization

Entities and Attributes (Normalized to 3NF)

1. PASSENGER

- Passenger_ID (PK, NUMBER)
- First_Name (VARCHAR2, NOT NULL)
- Last_Name (VARCHAR2, NOT NULL)
- Phone (VARCHAR2, UNIQUE, NOT NULL)
- Email (VARCHAR2, UNIQUE)
- Registration_Date (DATE, DEFAULT SYSDATE)

2. SMART_CARD

- Card_ID (PK, NUMBER)
- Passenger_ID (FK → PASSENGER, NUMBER, NOT NULL)
- Card_Number (VARCHAR2, UNIQUE, NOT NULL)
- Issue_Date (DATE, DEFAULT SYSDATE)
- Balance (NUMBER, DEFAULT 0, CHECK >= 0)
- Status (VARCHAR2, DEFAULT 'ACTIVE', CHECK IN ('ACTIVE', 'BLOCKED', 'EXPIRED'))

3. BUS

- Bus_ID (PK, NUMBER)
- Bus_Number (VARCHAR2, UNIQUE, NOT NULL)
- Capacity (NUMBER, CHECK > 0)
- Model (VARCHAR2, NOT NULL)
- Status (VARCHAR2, DEFAULT 'ACTIVE', CHECK IN ('ACTIVE', 'MAINTENANCE'))

4. DRIVER

- Driver_ID (PK, NUMBER)
- First_Name (VARCHAR2, NOT NULL)
- Last_Name (VARCHAR2, NOT NULL)
- License_Number (VARCHAR2, UNIQUE, NOT NULL)
- Phone (VARCHAR2, UNIQUE, NOT NULL)

5. ROUTE

- Route_ID (PK, NUMBER)
- Route_Name (VARCHAR2, NOT NULL)
- Start_Location (VARCHAR2, NOT NULL)
- End_Location (VARCHAR2, NOT NULL)
- Base_Fare (NUMBER, CHECK > 0)

6. SCHEDULE

- Schedule_ID (PK, NUMBER)

- Route_ID (FK → ROUTE, NUMBER, NOT NULL)
- Bus_ID (FK → BUS, NUMBER, NOT NULL)
- Driver_ID (FK → DRIVER, NUMBER, NOT NULL)
- Departure_Time (TIMESTAMP, NOT NULL)
- Arrival_Time (TIMESTAMP, NOT NULL)
- Schedule_Date (DATE, NOT NULL)
- *Constraint:* CHECK (Arrival_Time > Departure_Time)

7. TRANSACTION_RECORD

- Transaction_ID (PK, NUMBER)
- Card_ID (FK → SMART_CARD, NUMBER, NOT NULL)
- Schedule_ID (FK → SCHEDULE, NUMBER, NULLABLE)
- Amount (NUMBER, NOT NULL)
- Transaction_Date (TIMESTAMP, DEFAULT SYSTIMESTAMP)
- Transaction_Type (VARCHAR2, CHECK IN ('FARE_DEDUCTION', 'RECHARGE', 'REFUND'))

8. PUBLIC_HOLIDAY

- Holiday_ID (PK, NUMBER)
- Holiday_Name (VARCHAR2, NOT NULL)
- Holiday_Date (DATE, UNIQUE, NOT NULL)

9. AUDIT_LOG

- Log_ID (PK, NUMBER)
- Table_Name (VARCHAR2, NOT NULL)
- Action_Type (VARCHAR2, CHECK IN ('INSERT', 'UPDATE', 'DELETE'))
- User_Name (VARCHAR2, NOT NULL)
- Action_Date (TIMESTAMP, DEFAULT SYSTIMESTAMP)
- Old_Value (VARCHAR2)
- New_Value (VARCHAR2)

Normalization Justification:

- **1NF:** All attributes contain atomic, indivisible values.
- **2NF:** All tables have a single-column Primary Key, eliminating partial dependencies.
- **3NF:** No transitive dependencies exist (e.g., Bus details depend solely on Bus_ID , not on Route_ID or Driver_ID).